Chapter 34 — Cache Invalidation Strategy

> **Part IV — Offline-First & Synchronization** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 12 (Caching Strategies — the layers/TTL/ETag mechanics), Ch 3 (SSOT), Ch 32 (delta sync — server-driven freshness), Ch 41 (remote config, previewed). This chapter goes deep on the **hard half** of caching: *when and how to invalidate*.
>
> **Reference surface in StreamCart:** keeping the cached catalog, a product's detail, its category listing, and the user's cart consistent after writes, server changes, and new app releases — without serving stale data or over-fetching.

34.1 Problem Definition and Real-World Context

Ch 12 built the cache **layers** (HTTP / memory / Room) and basic freshness (TTL + ETag). This chapter tackles the famously hard part — **invalidation**: deciding *when* a cached value is no longer valid and *what else* must be invalidated with it. Getting this wrong is the source of the most confusing bugs:

- **Stale-after-write.** A user edits their profile; the profile screen updates but the cached "account header" elsewhere still shows the old name — because the write invalidated one cache entry but not the **dependent** ones.
- **Over-invalidation.** Clearing the entire cache on any change throws away valid data, causing refetch storms and jank.
- **TTL-only staleness.** Relying solely on time means data is stale for up to the TTL after a change the client *could* have known about (write or server push).
- **Cross-release corruption.** A new app version changes a DTO/entity shape, but old cached data (wrong shape) is read and mis-mapped — no **cache versioning**.
- **Cascade misses.** Updating a product should invalidate the product detail *and* every category/search listing that includes it — a **dependency graph** the naive code ignores.

Invalidation strategy defines the **triggers** (time, write-through, event/push, version), **scope** (single entry, tag/group, dependency cascade), and **coordination** across cache layers and app releases. It's what makes a cache *correct*, not just fast.

Three practical use cases

1. **E-commerce (StreamCart).** A price change (server push) invalidates the product detail *and* its listings; a user's wishlist write invalidates the wishlist count badge everywhere.
2. **OTT (Watch).** New episodes invalidate the show page and the "continue watching" row; a config change invalidates feature-gated UI.
3. **Social (Social).** A deleted post invalidates the feed, the author's profile count, and any cached thread — coordinated invalidation across entries.

Why this is architectural

Invalidation defines the **consistency contract of the cache**: which triggers cause invalidation, how far it cascades, and how caches stay coherent across layers and releases — the properties that decide whether users ever see stale or corrupt cached data.

34.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Invalidate on local writes (write-through/after-write), not just time.
F2	Invalidate dependent/related entries (cascade by relationship or tag).
F3	Support server-driven invalidation (push / version / delta, Ch 32).
F4	Version the cache so incompatible app releases don't read stale-shaped data.
F5	Invalidate the right scope (entry / group / all) — not over-broad.
F6	Keep layers (memory/DB/HTTP, Ch 12) coherent on invalidation.

Non-functional requirements

Attribute	Policy
Consistency	No read-after-write staleness; dependents stay coherent.
Efficiency	Minimal re-fetch; precise scope; no cache-clear storms.
Freshness	Server changes reflected promptly (push/version) where it matters.
Reliability	Cross-release safety (cache versioning).
Observability	Invalidation events + stale-serve rate visible.

Scope

- **In scope:** invalidation triggers (time/write/event/push/version), scope (entry/tag/cascade), cache versioning across releases, cross-layer coherence, server-driven invalidation, dependency graphs.
- **Out of scope:** the cache *layers/TTL/ETag* (Ch 12), delta-sync token mechanics (Ch 32), DB migrations (Ch 36), conflict resolution (Ch 33).

Assumptions & constraints

- Layered cache from Ch 12 (memory LRU + Room + HTTP). Backend can signal freshness (version bump / push / `must-revalidate`). `minSdk 24`.

34.3 System Design Fundamentals

- **Invalidation triggers (combine them):**
  - **Time (TTL):** simplest; data is "valid for N" (Ch 12). Bounds staleness but can't react to known changes.
  - **Write-through / after-write:** a local mutation updates or **invalidates** the cached entry immediately (and its dependents) — eliminates read-after-write staleness. The most important trigger for correctness.
  - **Event / server-push:** the server notifies the client (FCM, Ch 49; WebSocket, Ch 63) that an entity changed → invalidate/refetch. Near-real-time freshness.
  - **Version / delta:** the server bumps an entity/dataset version (or the client pulls deltas, Ch 32) → invalidate what changed.
- **Invalidation scope.** Precise scope avoids both staleness and waste:
  - **Single entry** (by key/id).
  - **Group / tag** — entries tagged with a label (e.g., `"category:shoes"`) invalidated together.
  - **Dependency cascade** — invalidating an entity invalidates derived views (a product → its listings, search results, counts). Model dependencies explicitly (tags or a dependency map) rather than guessing.
  - **All** — nuclear option; reserve for logout/account-switch (Ch 22/42) or cache- version mismatch.
- **Cache versioning (cross-release).** A `cacheVersion` key bumped when the cached **shape** changes (DTO/entity/mapping) invalidates all old-shaped data on app update — preventing mis-mapping stale data (complements DB migrations, Ch 36).
- **Cross-layer coherence.** On invalidation, clear the entry in **every** layer (memory derived from DB, Ch 12) so they can't diverge; the SSOT (Room) remains authoritative.
- **Stale-while-revalidate (SWR, Ch 12).** Serve stale immediately, invalidate + refetch in the background — good UX for non-critical data; never for freshness- critical values.

Core insight: **combine triggers (write-through for correctness, push/version for freshness, TTL as a backstop), invalidate the precise scope (entry/tag/cascade), keep layers coherent, and version the cache across releases** — invalidation is a coordination problem, not a single `evict()`.

34.4 High-Level Architecture


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Components & ownership

- `Invalidator` turns a trigger into a scoped invalidation across layers.
- **Dependency map / tags** define cascade scope.
- **Cache version** guards cross-release shape changes.
- **Layers (Ch 12)** are kept coherent; Room (SSOT) is authoritative.

Direction: trigger → invalidator → scope (entry/tag/cascade/all) → evict across layers → next read refetches.

34.5 Detailed Component Design



Component	Responsibility	Note
`Invalidator`	apply scoped invalidation across layers	write/event/version/all
dependency graph / tags	define cascade scope	product → listings/search/counts
`CacheVersion`	cross-release shape guard	bump on DTO/entity change
layer coherence	evict in memory+DB+HTTP	Room SSOT authoritative (Ch 12)

34.6 End-to-End Data Flow

Write-through with dependency cascade



Server-driven invalidation (push)



34.7 Complete Kotlin Implementation

Package: `com.streamcart.core.cache.invalidation`.

34.7.1 Tag/dependency-aware invalidator

```
class Invalidator @Inject constructor(
    private val memory: MemoryCache<String, *>,     // Ch 12
    private val dao: GenericDao,                     // Room SSOT
    private val httpCache: HttpCacheEvictor,         // OkHttp cache (Ch 12)
    private val deps: CacheDependencyGraph,
) {
    // Write-through: evict the entry + its dependents (cascade).
    suspend fun onWrite(entityId: String, type: String) {
        invalidateEntry(entityId)
        deps.dependentsOf(entityId).forEach { invalidateEntry(it) }   // cascade
    }

    // Server-driven: invalidate changed ids (from push/version/delta).
    suspend fun onServerChange(ids: List<String>) = ids.forEach { onWrite(it, "server") }

    suspend fun invalidateTag(tag: String) =
        deps.entriesWithTag(tag).forEach { invalidateEntry(it) }      // group invalidation

    suspend fun invalidateAll() {                                     // logout / version
mismatch
        memory.clear(); dao.clearAll(); httpCache.evictAll()
    }

    private suspend fun invalidateEntry(id: String) {
        memory.invalidate(id)                                         // memory layer
        // Room: don't delete the row; mark stale so the next read revalidates,
        // OR delete if purely a cache (choice per data). Here: mark stale.
        dao.markStale(id)
        httpCache.evict(urlForEntity(id))                            // HTTP layer
    }
}
```

34.7.2 Dependency graph (tags + relationships)

```
class CacheDependencyGraph @Inject constructor(private val dao: TagDao) {
    // A product depends on: its detail, listings that include it, search, category.
    suspend fun dependentsOf(entityId: String): Set<String> =
        dao.listingsContaining(entityId) + dao.searchKeysFor(entityId) +
dao.categoryOf(entityId)
    suspend fun entriesWithTag(tag: String): Set<String> = dao.keysWithTag(tag)
    suspend fun tagsOf(entityId: String): Set<String> = dao.tagsFor(entityId)
}
// Entries are tagged on insert, e.g. product p1 tagged ["category:shoes","brand:acme"].
```

34.7.3 Cache versioning (cross-release)

```
class CacheVersion @Inject constructor(private val store: DataStore<Preferences>) {
    private val CURRENT = 7                          // bump when cached shape changes
(DTO/entity/mapping)
    private val key = intPreferencesKey("cache_version")

    // On app start: if stored version != current, wipe caches (old shape unreadable).
    suspend fun ensureCompatible(invalidator: Invalidator) {
        val stored = store.data.first()[key] ?: 0
        if (stored != CURRENT) {
            invalidator.invalidateAll()             // old-shaped data gone (complements DB
migration, Ch 36)
            store.edit { it[key] = CURRENT }
        }
    }
}
```

34.7.4 Server-driven invalidation hook (push, Ch 49)

```
// FCM data message: {"invalidate": ["product:p1","category:shoes"]}
suspend fun onInvalidationPush(payload: InvalidationPush, invalidator: Invalidator) {
    payload.entries.forEach { ref ->
        when {
            ref.startsWith("product:") ->
invalidator.onServerChange(listOf(ref.removePrefix("product:")))
            ref.startsWith("category:") -> invalidator.invalidateTag(ref)
        }
    }
    // then trigger a delta sync (Ch 32) / WorkManager (Ch 30) to refetch
}
```

34.7.5 Module structure

```
:core:cache/invalidation  Invalidator, CacheDependencyGraph, CacheVersion
:core:cache               MemoryCache, HttpCacheEvictor (Ch 12)
:core:database            markStale, tag/relationship DAOs (Ch 3)
:core:notifications       invalidation push handler (Ch 49)
```

34.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Write + concurrent read of same entry	stale read after write	write-through evict in the same flow; Room SSOT re-emits
Cascade during active read	partial invalidation	evict in a transaction where atomicity matters; next read consistent
Push invalidation vs local pending write	clobber local edit	version-aware (Ch 33); don't evict un-synced local changes
Memory/DB divergence	stale layer	evict all layers together (Ch 12)
Invalidate-all vs in-flight refetch	wasted/incoherent	serialize; cancel in-flight; refetch after clear
Over-cascade (huge dep set)	refetch storm	bound cascade; tag precisely; SWR

- **Write-through eviction in the same code path as the write** is the core correctness rule: never let the write commit without invalidating the entry and its dependents, or read-after-write staleness returns.
- **Don't evict un-synced local changes** on a server push — the outbox (Ch 31) + conflict resolution (Ch 33) own those.

34.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Stale-after-write	QA/user report	add write-through cascade for that entity	coherent
Over-invalidation (storms)	refetch spikes	precise scope / tags; SWR	efficient
Cross-release mis-map	parse/display bug	bump cache version → wipe old-shaped data	correct
Missed dependent	stale derived view	add to dependency graph/tags	coherent
Push lost (offline)	—	TTL backstop + next sync (Ch 32)	eventual
Invalidate-all too often	perf	reserve for logout/version mismatch	bounded
Evicting pending local write	lost edit	exclude un-synced entries (Ch 31/33)	preserved


34.9 Failure Scenarios and Recovery

> **Precision matters both ways:** too-narrow invalidation leaves stale dependents; too-broad causes refetch storms. Model dependencies explicitly (tags/graph) so you invalidate exactly what changed and no more.

34.10 Scaling the Implementation

- **Level 1.** TTL + write-through evict of the single entry (Ch 12).
- **Level 2.** Dependency cascade (tags/graph) so dependents stay coherent; cross- layer eviction.
- **Level 3 — production.** Server-driven invalidation (push/version/delta); cache versioning across releases; SWR for non-critical.
- **Level 4 — large.** Fine-grained tag taxonomy; partial invalidation; remote- config invalidation rules (Ch 41); coordinated multi-cache coherence.
- **Level 5 — enterprise.** CDN/edge + client invalidation coordination, event- sourced invalidation, cache-coherence SLAs, observability on staleness.

Avoid overengineering: many apps need only TTL + write-through on single entries. Add dependency cascades and push invalidation when derived views and real-time freshness genuinely matter.

34.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Trigger	Write-through (after-write)	TTL only	Write-through for read-after-write correctness; TTL as a backstop
Freshness	Server push/version	poll/TTL	Push for near-real-time (Ch 49/63); poll for simplicity
Scope	Dependency cascade (tags/graph)	invalidate-all	Cascade for precision; all only for logout/version mismatch
Update	Invalidate (refetch next)	update-in-place (write-through update)	Update-in-place avoids a refetch when you have the new value; invalidate when you don't
Cross-release	Cache version bump	hope the shape is compatible	Version bump — mis-mapping stale data is a real bug
Stale UX	SWR	block-until-fresh	SWR non-critical; block freshness-critical (Ch 12)

Invalidate vs update-in-place: when a write already has the new value, **update** the cache in place (write-through update) — no refetch. When you only know *something changed* (server push without payload), **invalidate** and let the next read refetch. Prefer update-in-place where you have the data to avoid an extra round trip.

Cascade (tags/graph) vs invalidate-all: a dependency cascade keeps valid data and refetches only what's affected; invalidate-all is simple but throws away valid entries and causes storms. Model dependencies once (tags) and reap precise invalidation; reserve the nuclear option for logout/version changes.

34.12 Security Considerations

- **Invalidate on logout / account switch** (Ch 22/42) — stale cached data from a previous user is a leak; `invalidateAll()` on session end.
- **Don't trust server-push invalidation blindly:** a push says *what* changed, but the client must refetch over an authenticated, pinned channel (Ch 19/23) and the backend authorizes the refetch — a spoofed push shouldn't reveal data.
- **Cache version** also guards against reading data cached under a prior (possibly less-secure) scheme after an upgrade.
- **Don't cache (or promptly invalidate) sensitive/entitlement data** — a cached "premium=true" must not outlive a downgrade (invalidate on entitlement change).
- Client vs backend: invalidation keeps the client replica coherent; the backend remains authoritative and must re-validate on each refetch.

34.13 Performance and Resource Optimization

- **Precise invalidation** avoids refetch storms (bandwidth/battery) — the main perf lever here.
- **Update-in-place** over invalidate+refetch when you have the value.
- **SWR** hides latency for non-critical data.
- **Batch/coalesce** invalidations (e.g., a sync that changed 100 items → one pass).
- **Bound cascades** so one change can't invalidate the world.
- **Profiling:** invalidation frequency, refetch volume post-invalidation, stale-serve rate, cache hit-rate before/after (Ch 12).

34.14 Testing Strategy

Pure-ish invalidation logic + in-memory Room; assert coherence and scope.

```
class InvalidationTest {
    @Test fun `write-through evicts entry across layers`() = runTest {
        memory.put("p1", product);
        invalidator.onWrite("p1", "product")
        assertNull(memory.get("p1"))                // memory cleared
        assertTrue(dao.isStale("p1"))               // DB marked stale
    }

    @Test fun `cascade invalidates dependents`() = runTest {
        deps.link(product = "p1", listings = listOf("home", "category:shoes"))
        invalidator.onWrite("p1", "product")
        assertTrue(dao.isStale("home") && dao.isStale("category:shoes"))
    }

    @Test fun `tag invalidation clears group`() = runTest {
        dao.tag("p1", "category:shoes"); dao.tag("p2", "category:shoes")
        invalidator.invalidateTag("category:shoes")
        assertTrue(dao.isStale("p1") && dao.isStale("p2"))
    }

    @Test fun `cache version mismatch wipes all`() = runTest {
        store.set("cache_version", 6)               // old
        CacheVersion(store).ensureCompatible(invalidator)   // CURRENT=7
        assertTrue(memory.isEmpty() && dao.isEmpty())
    }

    @Test fun `server push does not evict unsynced local write`() = runTest {
        dao.upsert(product("p1", syncStatus = PENDING))
        invalidator.onServerChange(listOf("p1"))
        assertFalse(dao.isStale("p1"))              // local pending preserved (Ch 31/33)
    }
}
```

Also: logout invalidate-all, update-in-place vs invalidate, SWR behavior, no over-cascade.

34.15 Production Observability

- **Metrics:** invalidation events by trigger (write/push/version/TTL/all), refetch volume per invalidation, stale-serve rate, cache hit-rate (Ch 12), invalidate-all frequency, cache-version-wipe rate (per release).
- **Logs:** `invalidate{trigger, scope, count}`, `cache_version{from,to}` (no PII, Ch 28).
- **Debugging:** stale-after-write → missing write-through/cascade; refetch storms → over-broad invalidation; mis-mapped data after release → cache version not bumped; cross-account stale → not invalidating on logout; derived view stale → missing dependency.
- **Alerting:** stale-serve-rate spike or invalidate-all spike → coherence/perf issue.

- **Privacy:** don't log entity contents; counts/ids only.

34.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Why is invalidation "the hard part"?* Deciding *when* a value is invalid and *what else* depends on it.
2. *Invalidation triggers?* Time (TTL), write-through, server push/version, cache version (release).
3. *Why isn't TTL enough?* Read-after-write staleness; can't react to known changes.
4. *Write-through invalidation?* A write evicts/updates the entry (and dependents) immediately — correctness.
5. *Server-driven invalidation?* Push/version/delta tells the client what changed (Ch 49/32).

Scope/cascade 6. *Invalidation scope options?* Entry, tag/group, dependency cascade, all. 7. *Dependency cascade?* Invalidating an entity invalidates derived views (listings/ search/counts) via tags/graph. 8. *When invalidate-all?* Logout/account switch (Ch 22/42), cache-version mismatch. 9. *Over- vs under-invalidation?* Over → storms; under → stale dependents; model dependencies precisely.

Cross-release/layer 10. *Cache versioning?* Bump a version when the cached shape changes → wipe old data on upgrade (complements DB migration, Ch 36). 11. *Cross-layer coherence?* Evict in memory+DB+HTTP together; Room SSOT authoritative. 12. *Invalidate vs update-in-place?* Update when you have the value; invalidate when you only know it changed.

Correctness 13. *Push vs local pending write?* Don't evict un-synced local changes (Ch 31/33). 14. *SWR?* Serve stale, refetch in background (non-critical). 15. *Freshness-critical data?* Short TTL + invalidate on change; never SWR.

Security 16. *Logout?* Invalidate all (prevent cross-user leak). 17. *Trust a push?* Refetch over authenticated/pinned channel; backend authorizes. 18. *Entitlement data?* Invalidate on change (don't cache a stale "premium").

Performance 19. *Avoid storms?* Precise scope; update-in-place; batch; bound cascades. 20. *Measure?* Stale-serve rate, refetch volume, hit-rate.

Testing 21. *How test cascade?* Link dependents; write; assert all invalidated. 22. *How test cache version?* Mismatch → wipe. 23. *How test push-vs-local?* Push shouldn't evict pending local.

Production debugging 24. *Stale-after-write?* Missing write-through/cascade. 25. *Mis-mapped after release?* Cache version not bumped. 26. *Cross-account stale?* Not invalidating on logout.

Design-change challenges

1. **Profile name stale in the header after edit.** Write-through cascade to the header entry.
2. **Price change must reflect instantly.** Server push (Ch 49) → invalidate + delta.
3. **DTO shape changed in a release.** Bump cache version → wipe.
4. **One write invalidates too much.** Precise tags; bound cascade.

5. **Real-time collaborative data.** WebSocket push invalidation (Ch 63).

Mock walkthrough (abridged)

Clarify (what changes, dependents, real-time?) → triggers (write-through + push/version + TTL backstop) → scope (entry/tag/cascade; all for logout/version) → cross-layer coherence + cache versioning → don't evict pending local → security (logout, authorized refetch) → testing coherence.

34.17 Practical Exercises

1. **Beginner.** Add write-through invalidation of a single entry across memory+DB+ HTTP. *Eval:* no read-after-write staleness.
2. **Intermediate.** Add tag/dependency cascade so editing a product invalidates its listings/search. *Eval:* dependents coherent; no over-cascade.
3. **Advanced.** Add cache versioning (wipe on release shape change) + server-push invalidation. *Eval:* no mis-map after upgrade; push refreshes.
4. **Production debugging.** Fix a stale account header (missing cascade) and mis-mapped data after a release (no cache version). *Eval:* coherent; versioned.
5. **System design challenge.** Design precise invalidation that never evicts pending local writes and invalidates-all on logout. *Eval:* local preserved; no cross-user leak.

(Solutions withheld unless requested.)

34.18 Chapter Summary & Reference Sheet

Key decisions: combine **triggers** — **write-through** (read-after-write correctness), **server push/version** (freshness, Ch 49/32), **TTL** backstop (Ch 12), **cache version** (cross-release) ; invalidate the **precise scope** (entry / tag / dependency cascade), reserving **invalidate-all** for logout/account- switch/version-mismatch; keep **all cache layers coherent** (Room SSOT authoritative); **don't evict un-synced local writes** (Ch 31/33); prefer **update- in-place** when you have the value; **SWR** only for non-critical data. Invalidation is a **coordination** problem (dependencies + layers + releases), not an `evict()`.

Concepts: invalidation triggers (time/write/event/version), scope (entry/tag/ cascade/all), dependency graph/tags, cache versioning, cross-layer coherence, server-driven invalidation, update-in-place vs invalidate, SWR.

Common mistakes: TTL-only (stale-after-write); over-invalidation (storms); missing dependent cascade; no cache versioning (cross-release mis-map); evicting pending local writes; not invalidating on logout; inconsistent layers.

Checklist: ☐ write-through invalidate on mutation ☐ dependency cascade (tags/ graph) ☐ server push/version invalidation ☐ cache version bump per shape change ☐ evict all layers coherently ☐ invalidate-all on logout ☐ don't evict un-synced writes ☐ update-in-place when possible ☐ SWR for non-critical ☐ metrics on stale-serve/refetch.

Key APIs/patterns: write-through eviction, tag/dependency maps, `CacheVersion` (DataStore), `Invalidator` (entry/tag/cascade/all), OkHttp `Cache.evictAll`, Room `markStale`, push invalidation (Ch 49), delta (Ch 32).

References:

- (builds on) Ch 12 Caching Strategies
- HTTP caching / revalidation (MDN): https://developer.mozilla.org/docs/Web/HTTP/Caching
- Offline-first (freshness): https://developer.android.com/topic/architecture/data-layer/offline-first
- (composes) Ch 32 Delta Sync, Ch 33 Conflicts, Ch 36 Migrations, Ch 49 Push

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; invalidator + dependency graph + cache version + push hook; concurrency (write- through, don't-evict-pending); failure (stale/over-invalidation/mis-map); 5 levels; invalidate-vs-update and cascade-vs-all trade-offs; security (logout/authorized refetch); perf (precise scope); coherence tests; observability (stale-serve); 26 Q&A + 5 challenges. ✅
- **Expand later:** push (Ch 49), real-time (Ch 63), migrations (Ch 36), remote-config rules (Ch 41).
- **Interview coverage:** exceeds ≥25 bar.